

Ch 1 — Curves

Setup and arc length

Curve $\alpha : I \rightarrow \mathbb{R}^3$ *regular* (non-singular) iff $\alpha'(t) \neq \mathbf{0}$. Arc length: $s(t) = \int_{t_0}^t |\alpha'(u)| du$, so $s'(t) = |\alpha'(t)| > 0$.

Reparametrization. every regular curve admits arc-length param. $\alpha(s)$ with $|\alpha'(s)| = 1$ (uses Inverse Function Thm: $s' > 0 \Rightarrow t(s)$ smooth).

Frenet trihedron (arc length)

Assume $\alpha''(s) \neq \mathbf{0}$. Define

$$\mathbf{t} = \alpha'(s), \quad \mathbf{n} = \frac{\alpha''(s)}{|\alpha''(s)|}, \quad \mathbf{b} = \mathbf{t} \wedge \mathbf{n}.$$

$(\mathbf{t}, \mathbf{n}, \mathbf{b})$ is right-handed orthonormal. *Osculating plane* = $\text{span}(\mathbf{t}, \mathbf{n})$.

Lemma. if $|\mathbf{v}(t)|$ constant then $\mathbf{v} \cdot \mathbf{v}' = 0$. Used everywhere.

Curvature & torsion

$$k(s) = |\alpha''(s)| \geq 0, \quad \mathbf{b}'(s) = \tau(s)\mathbf{n}(s).$$

Radius of curvature $1/k$ = radius of *osculating circle* (best-fitting circle at $\mathbf{p}$).

$$\text{Frenet-Serret.} \quad \begin{pmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{pmatrix}' = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & -\tau \\ 0 & \tau & 0 \end{pmatrix} \begin{pmatrix} \mathbf{t} \\ \mathbf{n} \\ \mathbf{b} \end{pmatrix}$$

i.e. $\mathbf{t}' = k\mathbf{n}$, $\mathbf{n}' = -k\mathbf{t} - \tau\mathbf{b}$, $\mathbf{b}' = \tau\mathbf{n}$. Skew-symmetric.

General parametrization

$$\mathbf{t} = \frac{\alpha'}{|\alpha'|}, \quad \mathbf{b} = \frac{\alpha' \wedge \alpha''}{|\alpha' \wedge \alpha''|}, \quad \mathbf{n} = \mathbf{b} \wedge \mathbf{t}$$

$$k = \frac{|\alpha' \wedge \alpha''|}{|\alpha'|^3}, \quad \tau = -\frac{(\alpha' \wedge \alpha'') \cdot \alpha'''}{|\alpha' \wedge \alpha''|^2}.$$

Fundamental Theorem of Curves

(k, τ) determine $\alpha(s)$ up to rigid motion. Existence via Picard-Lindelöf on Frenet ODE (initial frame); uniqueness via $f(s) = |\mathbf{t}_1 - \mathbf{t}_2|^2 + |\mathbf{n}_1 - \mathbf{n}_2|^2 + |\mathbf{b}_1 - \mathbf{b}_2|^2$ has $f' \equiv 0$.

Rigid motion invariance. if $\beta = R\alpha + \mathbf{p}$ with $R \in \text{SO}(3)$, then $k_\beta = k_\alpha$, $\tau_\beta = \tau_\alpha$.

Standard examples

Curve	k	τ
Line	0	undef
Circle radius r	$1/r$	0
Helix $(a \cos t, a \sin t, bt)$	$\frac{a}{a^2+b^2}$	$\frac{b}{a^2+b^2}$
Catenary helix $(\cosh t, \sinh t, t)$	$k = \tau = \frac{1}{2 \cosh^2 t}$	

Ch 2 — Regular Surfaces

Definition (3 conditions)

$S \subset \mathbb{R}^3$ regular iff $\forall \mathbf{p} \in S$, $\exists$ chart $\mathbf{x} : U \subset \mathbb{R}^2 \rightarrow V \cap S$:

C1: $\mathbf{x}$ smooth. **C2:** $\mathbf{x}$ homeomorphism. **C3:** $d\mathbf{x}_q$ has rank

2 $\forall q \in U$ (equiv. $d\mathbf{x}_q$ injective; equiv. $\mathbf{x}_u \wedge \mathbf{x}_v \neq \mathbf{0}$).

Regularity equivalents. (i) rank 2; (ii) cols of $d\mathbf{x}$ lin. indep.;

(iii) $\mathbf{x}_u \wedge \mathbf{x}_v \neq \mathbf{0}$; (iv) some 2×2 minor $\frac{\partial(F_i, F_j)}{\partial(u, v)} \neq 0$.

Three classes of examples

• *Graphs.* $z = f(x, y)$ smooth $\Rightarrow$ regular surface, chart $\mathbf{x}(u, v) = (u, v, f(u, v))$.

• *Level sets.* $f^{-1}(a)$ regular if a is a *regular value*, i.e. $\nabla f \neq \mathbf{0}$ on $f^{-1}(a)$. (Inverse Function Thm.)

• *Surfaces of revolution.* $\mathbf{x}(u, v) = (f(v) \cos u, f(v) \sin u, g(v))$, $f > 0$. Then $d\mathbf{x} = \begin{pmatrix} -f \sin u & f' \cos u \\ f \cos u & f' \sin u \\ 0 & g' \end{pmatrix}$, rank 2 iff $(f', g') \neq (0, 0)$.

Cone with apex. $\{x^2 + y^2 = z^2\}$ *not* regular ($\nabla f = \mathbf{0}$ at apex).

Tangent plane (two equiv. defs)

(i) $T_{\mathbf{p}}S = \text{span}(\mathbf{x}_u, \mathbf{x}_v)$ (column space of $d\mathbf{x}_q$).

(ii) $T_{\mathbf{p}}S = \{\alpha'(0) : \alpha : (-\epsilon, \epsilon) \rightarrow S, \alpha(0) = \mathbf{p}\}$.

Equivalence (Sec 2.8): every $\mathbf{v} = a\mathbf{x}_u + b\mathbf{x}_v$ realized by $\alpha(t) = \mathbf{x}(u_0 + at, v_0 + bt)$.

Unit normal $\mathbf{N} = \frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}$.

For level set $\{f = 0\}$: $\nabla f \perp T_{\mathbf{p}}S$ (T11 Q2).

Smoothness; differential

$f : S \rightarrow \mathbb{R}$ smooth at $\mathbf{p}$ iff $f \circ \mathbf{x}$ smooth. $\phi : S_1 \rightarrow S_2$ smooth iff $\mathbf{y}^{-1} \circ \phi \circ \mathbf{x}$ smooth on overlap.

Differential. $d\phi_{\mathbf{p}}(\mathbf{v}) = (\phi \circ \alpha)'(0)$, $\alpha(0) = \mathbf{p}$, $\alpha'(0) = \mathbf{v}$. Linear, well-defined.

Change of params. transition $\Phi = \mathbf{y}^{-1} \circ \mathbf{x}$ is smooth diffeo.

First fundamental form

$I_{\mathbf{p}}(\mathbf{v}) = \langle \mathbf{v}, \mathbf{v} \rangle$. With $\mathbf{v} = a\mathbf{x}_u + b\mathbf{x}_v$:

$$E = \langle \mathbf{x}_u, \mathbf{x}_u \rangle, \quad F = \langle \mathbf{x}_u, \mathbf{x}_v \rangle, \quad G = \langle \mathbf{x}_v, \mathbf{x}_v \rangle$$

$$|\mathbf{v}|^2 = Ea^2 + 2Fab + Gb^2, \quad \cos \theta = \frac{\langle \mathbf{v}, \mathbf{w} \rangle}{|\mathbf{v}||\mathbf{w}|}.$$

Always: $EG - F^2 = |\mathbf{x}_u \wedge \mathbf{x}_v|^2 > 0$ (Lagrange identity).

Arc length: $s = \int \sqrt{Eu'^2 + 2Fu'v' + Gv'^2} dt$.

Area: $A = \iint \sqrt{EG - F^2} du dv$.

Polar in plane (T11 Q3): $\left| \frac{d\alpha}{d\theta} \right|^2 = r^2 + r'^2$.

Ch 3 — Curvatures of Surfaces

3.2 Linear algebra review

$(V, \langle \cdot, \cdot \rangle)$ 2-dim inner product space, $T : V \rightarrow V$ linear.

Self-adjoint. $\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T\mathbf{w} \rangle \forall \mathbf{v}, \mathbf{w}$.

Matrix rep. $T(\mathbf{x}_1) = a\mathbf{x}_1 + c\mathbf{x}_2$, $T(\mathbf{x}_2) = b\mathbf{x}_1 + d\mathbf{x}_2$ gives $[T]_{\mathcal{B}, \mathcal{B}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. For $V = \mathbb{R}^2$ with std inner product, $\langle A\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, A\mathbf{w} \rangle \iff A^\top = A$ ($b = c$).

Lemma 2. If $\mathcal{B}$ is *orthonormal*, then T self-adjoint $\iff [T]_{\mathcal{B}, \mathcal{B}}$ symmetric.

Char. poly. $p(\lambda) = \lambda^2 - (a + d)\lambda + (ad - bc)$, eigenvalues

$$\lambda_{1,2} = \frac{a+d}{2} \pm \sqrt{\left(\frac{a-d}{2}\right)^2 + b^2}.$$

Symmetric A : discriminant ≥ 0 , so $\lambda_1, \lambda_2 \in \mathbb{R}$.

Spectral Thm (Thm 3). Real symmetric 2×2 matrix $A = PDP^{-1}$ with $D = \text{diag}(\lambda_1, \lambda_2)$ real, P orthogonal ($P^\top = P^{-1}$). Eigenvectors of distinct eigenvalues are perpendicular.

Cor 4. Self-adjoint T admits orthonormal eigenbasis $\{\mathbf{v}_1, \mathbf{v}_2\}$.

3.3 The Gauss map

$N : S \rightarrow S^2$, $\mathbf{p} \mapsto \mathbf{N}(\mathbf{p})$ unit normal. Requires *orientability* (Möbius strip non-orientable, no Gauss map).

In a chart: $N(\mathbf{x}(u, v)) = \frac{\mathbf{x}_u \wedge \mathbf{x}_v}{|\mathbf{x}_u \wedge \mathbf{x}_v|}$, smooth in (u, v) .

Examples: Plane $ax + by + cz = 0$: N const, image = single pt. Sphere: $N(\mathbf{p}) = \mathbf{p}$, identity. Cylinder: image = equator of S^2 .

3.4 The map dN & second fundamental form

Identification $T_{N(\mathbf{p})}S^2 = T_{\mathbf{p}}S$ (both $\perp \mathbf{N}$); hence $dN_{\mathbf{p}} : T_{\mathbf{p}}S \rightarrow T_{\mathbf{p}}S$.

Self-adjointness of dN . differentiate $\langle \mathbf{N}, \mathbf{x}_u \rangle = 0$ in v and $\langle \mathbf{N}, \mathbf{x}_v \rangle = 0$ in u : $\langle N_v, \mathbf{x}_u \rangle = \langle N_u, \mathbf{x}_v \rangle$, hence $\langle dN(\mathbf{x}_v), \mathbf{x}_u \rangle = \langle dN(\mathbf{x}_u), \mathbf{x}_v \rangle$.

2nd fundamental form $\text{II}_{\mathbf{p}}(\mathbf{v}) = -\langle dN(\mathbf{v}), \mathbf{v} \rangle$. Two equivalent forms (via differentiating $\langle \mathbf{N}, \mathbf{x}_u \rangle = 0$ etc.):

$$e = -\langle N_u, \mathbf{x}_u \rangle = \langle \mathbf{N}, \mathbf{x}_{uu} \rangle,$$

$$f = -\langle N_u, \mathbf{x}_v \rangle = -\langle N_v, \mathbf{x}_u \rangle = \langle \mathbf{N}, \mathbf{x}_{uv} \rangle,$$

$$g = -\langle N_v, \mathbf{x}_v \rangle = \langle \mathbf{N}, \mathbf{x}_{vv} \rangle.$$

The $\langle \mathbf{N}, \mathbf{x}_{ij} \rangle$ form is usually easier.

3.5 Normal curvature; 3.6 principal/Gauss/mean

$\kappa_n(\mathbf{v}) = \text{II}_{\mathbf{p}}(\mathbf{v})$ for unit $\mathbf{v}$. *Geometric:* for unit-speed α on S with $\alpha'(0) = \mathbf{v}$, $\kappa_n = \langle \alpha''(0), \mathbf{N} \rangle = k \cos \theta$ where $\theta = \angle(\mathbf{n}, \mathbf{N})$.

Meusnier. all unit-speed curves through $\mathbf{p}$ with same tangent share same κ_n .

By Cor 4 applied to $-dN$: orthonormal eigenbasis $\{\mathbf{e}_1, \mathbf{e}_2\}$, eigenvalues $\kappa_1 \geq \kappa_2$ are the *principal curvatures* (= max/min of κ_n). Eigenvectors are *principal directions*. In this basis, $-[dN]_{\mathcal{B}} = \text{diag}(\kappa_1, \kappa_2)$.

Gauss & mean curvatures

$$K = \det(dN_{\mathbf{p}}) = \kappa_1 \kappa_2 = \frac{eg - f^2}{EG - F^2}$$

$$H = -\frac{1}{2} \text{tr}(dN_{\mathbf{p}}) = \frac{\kappa_1 + \kappa_2}{2} = \frac{eG - 2fF + gE}{2(EG - F^2)}.$$

Euler. $\mathbf{v} = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2 \Rightarrow \kappa_n(\theta) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta$. (Proof: in eigenbasis $\text{II}(\mathbf{v}) = \kappa_1 a^2 + \kappa_2 b^2$ with $a = \cos \theta, b = \sin \theta$.)

3.7 The map dN in local coords

Set $M(u, v) := N(\mathbf{x}(u, v)) = (N \circ \mathbf{x})(u, v)$. Decomposing dN in basis $\{\mathbf{x}_u, \mathbf{x}_v\}$:

$$[dN]_{\mathcal{B}} = -\left(\begin{smallmatrix} E & F \\ F & G \end{smallmatrix}\right)^{-1} \left(\begin{smallmatrix} e & f \\ f & g \end{smallmatrix}\right).$$

Recover $K = \det(dN)$, $H = -\frac{1}{2}\text{tr}(dN)$. *Note:* $[dN]_{\mathcal{B}}$ not symmetric in general (only when basis orthonormal).

3.8 worked example: $z = y^2 - x^2$

At origin: $E = G = 1, F = 0, e = -2, f = 0, g = 2 \Rightarrow K = -4 < 0$ (hyperbolic), $H = 0, \kappa_1 = -2, \kappa_2 = 2$, principal directions $\mathbf{x}_u, \mathbf{x}_v$.

Sign of K — local shape

Elliptic $K > 0$ (sphere-ish), **parabolic** $K = 0$ (cylinder-ish), **hyperbolic** $K < 0$ (saddle-ish).

Special points/curves

- *Umbilic*: $\kappa_1 = \kappa_2$ (so $dN = \lambda \text{Id}$). Connected all-umbilic $\Rightarrow$ plane or sphere.
- *Line of curvature*: α' always principal direction. **Rodrigues**. $N'(s) = -\kappa_i \alpha'(s)$.
- *Asymptotic curve*: $\text{II}(\alpha') = 0$ (exists only at $K \leq 0$).
- *Minimal surface*: $H \equiv 0 \iff \kappa_1 = -\kappa_2$. Examples: plane, catenoid, helicoid.

Standard examples

Surface	K	H	note
Plane	0	0	flat
Cylinder r	0	$1/(2r)$	developable
Sphere R	$1/R^2$	$1/R$	all umbilic
Catenoid	$-1/\cosh^4 v$	0	minimal
Helicoid	same as cat.	0	minimal
Pseudosphere	-1	—	const K

Ch 4 — Isometries & Theorema Egregium

Isometry, local isometry

Isometry. diffeo $\phi : S_1 \rightarrow S_2$ with $\langle d\phi(\mathbf{v}), d\phi(\mathbf{w}) \rangle = \langle \mathbf{v}, \mathbf{w} \rangle \forall \mathbf{v}, \mathbf{w}$. Equiv. $|d\phi(\mathbf{v})| = |\mathbf{v}|$ (lengths preserved); equiv. $\text{I}_{\mathbf{p}}(\mathbf{v}) = \text{I}_{\phi(\mathbf{p})}(d\phi(\mathbf{v}))$.

Local isometry at $\mathbf{p}$. isometry on some open $V_1 \ni \mathbf{p}$. *Local isometry* = at every point.

Warning. local isom. need NOT be bijection (e.g. plane $\rightarrow$ cylinder wraps).

Preserved by isometries: arc length, angles, area, K , geodesics, $\kappa_g, \Gamma_{ij}^k, \text{I}$.

First fund. form criterion (Prop 1). Charts $\mathbf{x}: U \rightarrow S_1, \mathbf{y}: U \rightarrow S_2$ on the same U . If $E_1 = E_2, F_1 = F_2, G_1 = G_2$ as functions on U , then $\Phi = \mathbf{y} \circ \mathbf{x}^{-1}: \mathbf{x}(U) \rightarrow \mathbf{y}(U)$ is an isometry.

Conformal maps

ϕ conformal iff $\langle d\phi\mathbf{v}, d\phi\mathbf{w} \rangle = \lambda^2 \langle \mathbf{v}, \mathbf{w} \rangle$ (preserves angles, scales lengths by λ). Equiv. $E_1 = \lambda^2 E_2, F_1 = \lambda^2 F_2, G_1 = \lambda^2 G_2$.

Isometry $\Leftrightarrow$ conformal with $\lambda \equiv 1$. Stereographic proj. $S^2 \setminus \{N\} \rightarrow \mathbb{R}^2$ is conformal but not isometry.

Christoffel symbols

Decompose $\mathbf{x}_{uu}, \mathbf{x}_{uv}, \mathbf{x}_{vv}$ in basis $\{\mathbf{x}_u, \mathbf{x}_v, \mathbf{N}\}$:

$$\mathbf{x}_{uu} = \Gamma_{11}^1 \mathbf{x}_u + \Gamma_{11}^2 \mathbf{x}_v + e\mathbf{N}, \text{ etc.}$$

Tangential coefficients Γ_{ij}^k (symmetric in lower indices: $\Gamma_{ij}^k = \Gamma_{ji}^k$).

Closed form (intrinsic!):

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell} (\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}),$$

$$(g_{ij}) = \left(\begin{smallmatrix} E & F \\ F & G \end{smallmatrix}\right), (g^{ij}) \text{ inverse.}$$

Theorema Egregium (Gauss 1827). K depends only on E, F, G and their derivatives. Hence isometries preserve K .

Brioschi formula (case $F = 0$)

$$K = -\frac{1}{2\sqrt{EG}} \left(\partial_u \frac{G_u}{\sqrt{EG}} + \partial_v \frac{E_v}{\sqrt{EG}} \right).$$

Verify on sphere: $E = 1, G = \sin^2 u \Rightarrow K = 1$.

Ch 5 — Vector Fields, Geodesics

Vector fields

$\mathbf{w}(\mathbf{p}) \in T_{\mathbf{p}}S$ smooth assignment. In chart: $\mathbf{w} = a(u, v)\mathbf{x}_u + b(u, v)\mathbf{x}_v$ with a, b smooth (chart-independent).

Hairy ball: every smooth tangent field on S^2 vanishes somewhere.

Covariant derivative

For $\mathbf{w}(t)$ along curve $\alpha(t)$ on S :

$$\frac{D}{dt} \mathbf{w} = \frac{d\mathbf{w}}{dt} - \left\langle \frac{d\mathbf{w}}{dt}, \mathbf{N} \right\rangle \mathbf{N}$$

i.e. orthogonal projection of $d\mathbf{w}/dt$ onto $T_{\mathbf{p}}S$.

Also $\frac{D\mathbf{w}}{du} = \partial_u \mathbf{w} - \langle \partial_u \mathbf{w}, \mathbf{N} \rangle \mathbf{N}$, similarly for v .

Properties: (a) linearity (b) product rule $\frac{D}{dt}(f\mathbf{v}) = f'\mathbf{v} + f\frac{D}{dt}\mathbf{v}$ (c) **metric compatibility** $\frac{d}{dt}\langle \mathbf{v}, \mathbf{w} \rangle = \langle \frac{D}{dt}\mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{v}, \frac{D}{dt}\mathbf{w} \rangle$.

In coords ($\mathbf{w} = A\mathbf{x}_u + B\mathbf{x}_v, \alpha = (u(t), v(t))$):

$$\frac{D}{dt} \mathbf{w} = (A' + \Gamma_{ij}^1 \dots) \mathbf{x}_u + (B' + \Gamma_{ij}^2 \dots) \mathbf{x}_v,$$

intrinsic (only Γ 's appear).

Compatible with isometry: $\frac{D}{dt}(d\phi(\mathbf{w})) = d\phi(\frac{D}{dt}\mathbf{w})$.

Parallel transport

$\mathbf{w}$ parallel along α iff $\frac{D}{dt}\mathbf{w} = \mathbf{0}$. Linear ODE $\Rightarrow$ unique parallel extension.

Two parallel fields $\mathbf{v}, \mathbf{w}$: $\langle \mathbf{v}(t), \mathbf{w}(t) \rangle$ const. So $|\mathbf{v}|$ const, angles preserved $\Rightarrow$ parallel transport is an orthogonal isomorphism between tangent planes (T9/10 Q4).

Geodesic curvature

Unit-speed $\alpha(s), \mathbf{s} := \mathbf{N} \wedge \mathbf{t}$. $\alpha'' \perp \mathbf{t}$, so:

$$\alpha''(s) = k_n \mathbf{N} + k_g \mathbf{s}, \quad k^2 = k_n^2 + k_g^2$$

$k_n = \langle \mathbf{N}, \alpha'' \rangle$ (extrinsic, $= \text{II}(\mathbf{t})$), $k_g = \langle \mathbf{s}, \alpha'' \rangle = \langle \mathbf{N} \wedge \mathbf{t}, \alpha'' \rangle$ (intrinsic).

$\frac{D}{ds} \alpha' = \text{orthog. projection of } \alpha'' \text{ onto } T_{\mathbf{p}}S$ has length $|k_g|$.

Geodesics (4 equiv. defs)

For arc-length-param α : TFAE
(i) $\frac{D}{ds} \alpha' = \mathbf{0}$ (ii) $\alpha''(s) \perp T_{\alpha(s)}S$
(iii) $\alpha'' \parallel \mathbf{N}$ (iv) $k_g \equiv 0$.

Geodesic ODE & existence

$\alpha = (u(t), v(t))$:

$$u'' + \Gamma_{11}^1 u'^2 + 2\Gamma_{12}^1 u'v' + \Gamma_{22}^1 v'^2 = 0,$$

$$v'' + \Gamma_{11}^2 u'^2 + 2\Gamma_{12}^2 u'v' + \Gamma_{22}^2 v'^2 = 0.$$

Existence/uniqueness. given $\mathbf{p} \in S$, unit $\mathbf{v} \in T_{\mathbf{p}}S$, $\exists!$ geodesic with $\alpha(0) = \mathbf{p}, \alpha'(0) = \mathbf{v}$.

Constant speed. $\frac{d}{dt} |\alpha'|^2 = 2\langle \alpha'', \alpha' \rangle = 0$. If $|\alpha'| = c$ const, reparametrize $\gamma(s) := \alpha(s/c)$ to unit speed.

Isometry-preservation

Local isom. sends geodesics to geodesics (k_g intrinsic). Used to find geodesics on cylinder/cone via unrolling.

Standard geodesics

Surface	geodesics
Plane	straight lines
Cylinder	vert. lines, horiz. circles, helices
Cone (no apex)	images of straight lines on unrolled
Sphere	great circles <i>only</i>
Flat torus	images of straight lines

Local minimization (5.8)

For $\mathbf{p}, \mathbf{q} \in S$ sufficiently close: $\exists!$ geodesic α from $\mathbf{p}$ to $\mathbf{q}$, and any other curve β has $L(\beta) \geq L(\alpha)$. *Locally* shortest, not globally.

Exponential map; Gauss lemma (5.9)

$\exp_{\mathbf{p}}: T_{\mathbf{p}}S \rightarrow S, \mathbf{v} \mapsto \gamma_{\mathbf{v}}(1)$ with $\gamma_{\mathbf{v}}(0) = \mathbf{p}, \gamma'_{\mathbf{v}}(0) = \mathbf{v}$. Local diffeo near $\mathbf{0}$ since $d(\exp_{\mathbf{p}})_0 = \text{Id}$. *Geodesic polar coords* (ρ, θ) :

$$E = 1, F = 0, ds^2 = d\rho^2 + G(\rho, \theta) d\theta^2$$

Gauss lemma. radial geodesics $\perp$ geodesic circles. Underlies length-minimization proof.

Variational viewpoint (5.10)

Geodesics = critical points of $L(\alpha) = \int |\alpha'| dt$ (or energy $E = \frac{1}{2} \int |\alpha'|^2 dt$). Euler–Lagrange = geodesic ODE.